Finite-Sample Confidence Sets for Nonsmooth Convex-to-Concave Transitions

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Abstract

Many scientific curves change from increasing returns to diminishing returns. A point estimate can mark this transition, but it does not show which other locations remain compatible with the data. The problem is difficult when the mean curve has a kink, a jump, or a flat transition region, because smooth derivative and bootstrap arguments can then fail. We introduce shape-contrast inversion (SCI), a finite-sample uncertainty layer for this problem. SCI builds simultaneous bounds for a fixed family of local chord contrasts and inverts only their certified signs. The output is an outer confidence set for all locations at which the mean curve can change from convex to concave. In fixed-design Gaussian regression, coverage is uniform and does not require a derivative, continuity at the transition, or a unique transition. We also give finite-sample versions for unknown constant noise, bounded heteroskedasticity, and independent replicate curves with arbitrary within-curve covariance. In 80,000 fresh simulations, known-scale coverage ranged from 97.70% to 98.04% at a 95% nominal level across 16 published S-shaped benchmark settings. In a matched comparison with a generic honest confidence-region projection, SCI reduced median width by 19.4%–75.7% for cusp, onset, and jump signals. Both methods were wide for a weak smooth logistic signal. Analyses of atmospheric LIDAR and 11 replicated DNase assay runs illustrate how the answer changes with the available noise information. A matrix-free implementation evaluates about six million contrasts for one million observations in less than one second on the benchmark machine.

Keywords: shape constraints; inflection location; finite-sample inference; partial identification; multiscale contrasts; S-shaped regression.

1 Introduction

Researchers often need to locate a change from acceleration to saturation. Examples include dose-response curves, biological growth, atmospheric profiles, disease progression, and production curves. Shape-constrained regression can estimate such a curve without choosing a detailed parametric model. In particular, an S-shaped fit is convex before a transition and concave after it [Feng et al., 2022]. The fitted transition is useful, but it does not answer a basic uncertainty question:

> Which transition locations can the data rule out while remaining valid for kinks, jumps, and flat transition regions?

This is a broadly relevant data-science problem. A reported change point is often treated as a physical threshold or an operating decision. If its uncertainty method silently assumes a smooth and unique inflection, a narrow interval can be more misleading than no interval. The difficulty is not only computational. When curvature is zero over an interval, the data-generating curve itself does not identify one preferred transition point. A valid method must then report a set rather than invent a unique target.

Earlier work already gives important forms of inference for qualitative features. Davies et al. [2009] construct finite-sample confidence regions for a regression curve and study features including inflection points. Schmidt-Hieber et al. [2013] use multiscale sign statements to locate roots of operators, including curvature changes. Shape-restricted splines and bootstrap methods provide practical point and interval estimators under smoother models [Meyer, 2008, Liao and Meyer, 2017]. These results show that the general topic is established and important. Our goal is therefore not to claim the first confidence method for an inflection point.

We address a narrower gap. We want a direct, computable, finite-sample confidence set for the full set of convex-to-concave transition locations. The validity statement should survive nonsmooth curves, nonunique transitions, and irregular fixed designs. The output should also be usable as an uncertainty layer around a modern S-shaped point estimate.

Our method, shape-contrast inversion (SCI), starts from local chord inequalities. A positive chord residual supports convexity on its interval; a negative residual supports concavity. We construct simultaneous confidence bounds for a fixed collection of averaged residuals. Certified positive signs exclude transition locations to their left, while certified negative signs exclude locations to their right. Inverting these statements gives an outer confidence set for every compatible transition location.

The contributions are as follows.

1. We define the target as the full identified set of convex-to-concave transitions. This makes flat transition regions explicit.
2. We give a direct finite-sample confidence set in fixed-design Gaussian regression. The theorem permits kinks, jumps, and nonunique transitions.
3. We extend the construction to unknown constant variance, a declared upper bound on variance heterogeneity, and independent replicate curves with arbitrary within-curve covariance.
4. We show localization under an explicit contrast-margin condition. The cubic special case has the familiar $(\log n/n)^{1/7}$ order; we do not claim that this order is new.
5. We give a matrix-free implementation and compare SCI with both a published smooth-bootstrap method and a matched honest projection baseline.
6. We report weak-information cases as well as successes. A wide set is an intended answer when the data cannot support precise localization.

The rest of the paper is organized as follows. Section 2 positions the method. Sections 3–5 define SCI and prove its main guarantee. Section 6 gives the noise extensions. Sections 8 and 9 report simulations and real-data examples. Section 10 states the limits and the remaining open theory.

2 Relation to earlier work

Confidence regions and multiscale inference. The confidence-region approach of Davies et al. [2009] is broad: it first constructs a set of plausible regression functions and then studies their qualitative features. Multiscale tests similarly turn many local statements into global qualitative conclusions [Dümbgen and Spokoiny, 2001, Schmidt-Hieber et al., 2013]. SCI follows the same honest-inference principle, but it targets one feature directly. It stores only the endpoints, scale, and sufficient sums needed for local chord contrasts. This direct inversion is the source of its simple proof and its computational scale.

S-shaped point estimation. The tuning-free least-squares estimator of Feng et al. [2022] estimates a monotone S-shaped curve and its transition. We use its implementation as the main point estimator in our hybrid workflow. SCI does not replace that fit. It supplies a confidence set around the scientific transition target. Our coverage theorem needs the convex-to-concave restriction but not monotonicity.

Spline change-point intervals. Liao and Meyer [2017] estimate a change point with shape-restricted regression splines. Their `ShapeChange` software includes a residual bootstrap interval. It is a useful comparator in its smooth spline regime. Our nonsmooth experiments deliberately include settings outside that regime, so those comparisons measure robustness rather than a failure of the published theory.

What is new here. The main object is a finite-sample outer confidence set for a possibly set-valued transition under a nonsmooth shape class. The most informative comparison is therefore not only against a point or smooth-bootstrap method, but also against another procedure with a matched finite-sample guarantee. For this purpose we implement a pointwise-band projection (PBP) baseline. It is related to the confidence-region strategy of Davies et al. [2009], but it is our deliberately simple baseline, not their official algorithm.

3 Model and target

Let

$$A < x_1 < \cdots < x_n < B, \qquad Y_i = f(x_i) + \varepsilon_i, \quad i = 1, \ldots, n,$$

where the design is fixed. In the basic model, $\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$ and σ is known. Extensions relax the scale assumption.

Definition 3.1 (Compatible transition set)**.** For a real-valued function f on $[A, B]$, define

$$\mathcal{I}(f) = \{m \in [A, B] : f \text{ is convex on } [A, m] \text{ and concave on } [m, B]\}.$$

We call every $m \in \mathcal{I}(f)$ a compatible convex-to-concave transition.

The definition does not require a derivative. It permits an endpoint transition, a kink, and a one-sided jump at the transition. It also permits a flat-curvature interval, in which case more than one m is compatible.

Lemma 3.2 (Geometry of the target)**.** *If $m_1 < m_2$ both belong to $\mathcal{I}(f)$, then f is affine on $[m_1, m_2]$ and $[m_1, m_2] \subseteq \mathcal{I}(f)$. Therefore, whenever $\mathcal{I}(f)$ is nonempty, it is an interval, possibly a singleton.*

This lemma separates statistical uncertainty from identification. Even with no noise, a curve that is affine over $[m_1, m_2]$ does not select one transition inside that interval. Our confidence statement therefore covers the whole set $\mathcal{I}(f)$.

3.1 Chord contrasts

For design indices $L < M < R$, define the chord residual

$$Q_{L,M,R}(f) = \frac{x_R - x_M}{x_R - x_L} f(x_L) + \frac{x_M - x_L}{x_R - x_L} f(x_R) - f(x_M). \tag{1}$$

Convexity makes (1) nonnegative; concavity makes it nonpositive. Single triples are noisy, so we average nonnegative collections of chord residuals. Let $\mathcal{T}$ be a fixed finite family. Each $T \in \mathcal{T}$ defines a row vector $w_T \in \mathbb{R}^n$, a support interval $[a_T, b_T]$, and a contrast mean

$$Z_T = w_T^\top Y, \qquad \mu_T = w_T^\top f(x_{1:n}).$$

The weights are chosen so that $\mu_T \geq 0$ when f is convex on the support, and $\mu_T \leq 0$ when f is concave there. The family is fixed before seeing Y. Appendix A gives the construction used in the experiments.

4 Shape-contrast inversion

Let $M = |\mathcal{T}|$ and let Φ be the standard normal distribution function. For a desired error probability α, set

$$c_{\alpha,M} = \Phi^{-1}\left(1 - \frac{\alpha}{2M}\right). \tag{2}$$

For every contrast, form the simultaneous interval

$$[\ell_T, u_T] = \left[Z_T - c_{\alpha,M}\sigma \left\|w_T\right\|_2, Z_T + c_{\alpha,M}\sigma \left\|w_T\right\|_2\right]. \tag{3}$$

A contrast is certified positive when $\ell_T > 0$ and certified negative when $u_T < 0$. Define

$$\widehat{L} = \max(\{A\} \cup \{a_T : \ell_T > 0\}), \tag{4}$$

$$\widehat{U} = \min(\{B\} \cup \{b_T : u_T < 0\}). \tag{5}$$

The SCI output is

$$\widehat{\mathcal{C}}(Y) = \begin{cases} [\widehat{L}, \widehat{U}], & \widehat{L} \leq \widehat{U}, \\ \varnothing, & \widehat{L} > \widehat{U}. \end{cases} \tag{6}$$

The construction has a simple interpretation. A reliably convex interval cannot lie wholly after a valid transition, so its left endpoint gives a lower limit. A reliably concave interval cannot lie wholly before a valid transition, so its right endpoint gives an upper limit. Only simultaneous sign statements are inverted.

Figure 1 shows the full construction on one deterministic example. The first and last panels use the released implementation; the chord panel explains the geometric sign rule. The figure is an explanation of the algorithm, not evidence for coverage. The coverage claim comes from Theorem 5.1.

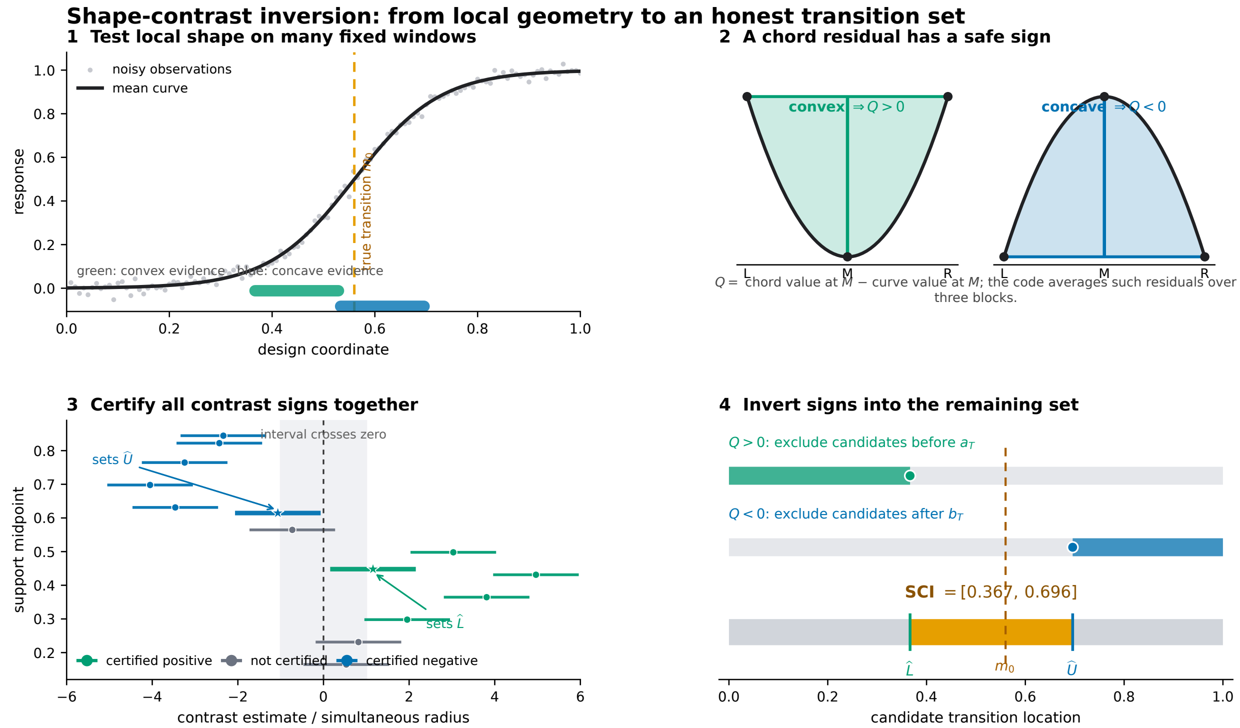

Figure 1: An executable overview of SCI. (1) A fixed multiscale family tests local windows of the noisy curve. The two highlighted windows determine the reported endpoints in this example. (2) A chord lies above a convex curve and below a concave curve, so the chord residual has a known sign. The implementation averages these residuals over three blocks. (3) Simultaneous intervals certify positive and negative contrast means; intervals that cross zero make no claim. Stars mark the two active contrasts. (4) Positive signs exclude candidate transitions on the left, negative signs exclude candidates on the right, and the unexcluded interval is the SCI confidence set.

5 Finite-sample theory

Theorem 5.1 (Known-scale finite-sample coverage). *Suppose $\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$, the finite contrast family $\mathcal{T}$ is fixed independently of Y, and every row has the sign property stated above. Then, uniformly over every f with $\mathcal{I}(f) \neq \varnothing$,*

$$\mathop{\mathrm{P}}_{f}\{\mathcal{I}(f) \subseteq \widehat{\mathcal{C}}(Y)\} \geq 1 - \alpha.$$

Consequently, $\mathrm{P}_f\{\widehat{\mathcal{C}}(Y) = \varnothing\} \leq \alpha$ whenever the target is nonempty.

The theorem is exact for the stated Gaussian model. It is not an asymptotic bootstrap result. It covers all compatible locations at once and therefore remains meaningful when the target is an interval.

Proposition 5.2 (Minimal interval from the certified signs). *Condition on the collection of certified positive and negative signs. Among closed intervals that use only those logical sign implications, $[\widehat{L}, \widehat{U}]$ is the smallest one containing every transition that has not been excluded.*

Proposition 5.2 is a logical minimality result. It does not say that SCI is the shortest possible confidence procedure. A different joint band or additional model assumptions can give a smaller set.

5.1 Localization under a contrast margin

Coverage alone does not ensure that the set is informative. The next result states the signal strength needed for localization. It also makes clear why weak smooth transitions can produce wide sets.

Assumption 5.3 (Contrast margin at scale d)**.** Write $\mathcal{I}(f) = [m_-, m_+]$. For a scale $d > 0$, suppose the family contains

(i) a left contrast T_- with $a_{T_-} \geq m_- - Kd$, $\mu_{T_-} \geq Bd^\gamma$, and $\|w_{T_-}\|_2 \leq C_w/\sqrt{nd}$; and

(ii) a right contrast T_+ with $b_{T_+} \leq m_+ + Kd$, $\mu_{T_+} \leq -Bd^\gamma$, and $\|w_{T_+}\|_2 \leq C_w/\sqrt{nd}$.

Here $B, C_w, K > 0$ and $\gamma > 0$ do not depend on n.

Theorem 5.4 (Margin-based localization)**.** *Under Theorem 5.1 and Assumption 5.3, let $0 < \eta < 1$ and suppose*

$$Bd^\gamma \geq \frac{\sigma C_w}{\sqrt{nd}}\left(c_{\alpha,M} + z_{1-\eta/2}\right), \tag{7}$$

where $z_q = \Phi^{-1}(q)$. Then, with probability at least $1 - \alpha - \eta$,

$$\mathcal{I}(f) \subseteq \widehat{\mathcal{C}}(Y) \subseteq [m_- - Kd, m_+ + Kd].$$

In particular, $|\widehat{\mathcal{C}}(Y)| \leq |\mathcal{I}(f)| + 2Kd$ on this event.

Balancing the two sides of (7) gives the detection scale

$$d_n \asymp \left\{\frac{\sigma^2(c_{\alpha,M} + z_{1-\eta/2})^2}{B^2 n}\right\}^{1/(2\gamma+1)}. \tag{8}$$

When M grows polynomially with n, $c^2_{\alpha,M} = O(\log n)$, so the excess length is conditionally $O_{\mathrm{P}}\{(\sigma^2 \log n/(B^2 n))^{1/(2\gamma+1)}\}$. This statement is conditional on the explicit margin and contrast availability. It is not yet a broad minimax theorem for every S-shaped function class.

5.2 Cubic transition

The standard smooth reference case is a cubic crossing. Let $x_i = i/(n+1)$ and

$$f(x) = c_0 + c_1(x - m_0) - B(x - m_0)^3, \qquad B > 0.$$

For the dyadic contrast family in Appendix A, define

$$h_* = \left\{\frac{\sigma(c_{\alpha,M_n} + z_{1-\eta/2})}{\sqrt{6}B\sqrt{n+1}}\right\}^{2/7}. \tag{9}$$

Corollary 5.5 (Exact cubic localization order)**.** *Put $q_* = (n+1)h_*$, assume $q_* \geq 1$, and let q be the smallest dyadic integer with $q \geq q_*$. If $3q \leq n$ and $m_0 \in [8h_*, 1 - 8h_*]$, then*

$$\mathop{\mathrm{P}}_f\{m_0 \in \widehat{\mathcal{C}}(Y),\ |\widehat{\mathcal{C}}(Y)| < 16h_*\} \geq 1 - \alpha - \eta.$$

Thus

$$|\widehat{\mathcal{C}}(Y)| = O_{\mathrm{P}}\left\{\left(\frac{\sigma^2 \log n}{B^2 n}\right)^{1/7}\right\}.$$

For the three frozen separation multipliers, $M_n \le 6n + 6(1 + \log_2 n)$. Hence the polynomial-size condition used for the displayed stochastic order holds directly.

The $n^{-1/7}$ power is the known regular order for localizing a root of a second derivative. Related orders appear in Schmidt-Hieber et al. [2013] and Feng et al. [2022]. The value of Corollary 5.5 is compatibility: SCI keeps finite-sample validity over a wider nonsmooth class without losing this reference order in the cubic case.

6 Unknown and nonconstant noise

The inversion only needs simultaneous bounds for the contrast means. This modularity gives three useful extensions.

6.1 Unknown constant scale

Let $X \in \mathbb{R}^{n\times r}$ be a fixed full-rank nuisance design and let $R_X = I - P_X$ be its residual projector. Write $q_{\nu,\eta}$ for the lower η quantile of a chi-squared random variable with ν degrees of freedom, and define

$$\overline{\sigma} = \left\{ \frac{\|R_X Y\|_2^2}{q_{n-r,\eta}} \right\}^{1/2}. \tag{10}$$

Theorem 6.1 (Honest upper scale bound). *If $Y = \mu + \varepsilon$ with arbitrary deterministic $\mu \in \mathbb{R}^n$ and $\varepsilon \sim N(0, \sigma^2 I)$, then*

$$\inf_{\mu,\sigma>0} \mathrm{P}_{\mu,\sigma}(\overline{\sigma} \ge \sigma) \ge 1 - \eta.$$

If (3) uses $\overline{\sigma}$ and error budget $\alpha - \eta$, the resulting SCI *set covers $\mathcal{I}(f)$ with probability at least $1 - \alpha$.*

No independence between the scale estimate and the contrasts is required. The price is conservatism: the residual norm can contain unremoved signal. Our default nuisance space uses fixed blocks of size two. For sampled monotone curves with bounded total variation, its signal contribution becomes negligible in the usual increasing-design limit.

6.2 Bounded heteroskedasticity

Now let the errors be independent with $\varepsilon_i \sim N(0, \sigma_i^2)$. A finite-sample result requires some information about the unobserved variance pattern. We make that information explicit.

Assumption 6.2 (Variance-ratio bound)**.** For a declared $\kappa \ge 1$,

$$\max_i \sigma_i^2 \le \kappa \bar{\sigma}^2, \qquad \bar{\sigma}^2 = \frac{1}{n}\sum_{i=1}^n \sigma_i^2.$$

Use a fixed partition into blocks of size two or three, and let R remove the constant within each block. Set $Q = \|RY\|_2^2$ and

$$d_{n,\kappa,\eta} = 1 - 2\sqrt{\frac{2\kappa \log(1/\eta)}{n}}, \tag{11}$$

$$\widehat{\sigma}^2_{\max} = \frac{2\kappa Q}{n d_{n,\kappa,\eta}}. \tag{12}$$

Theorem 6.3 (Heteroskedastic scale envelope)**.** *Under Assumption 6.2, if $d_{n,\kappa,\eta} > 0$, then*

$$\mathrm{P}\left(\widehat{\sigma}^2_{\max} \geq \max_i \sigma_i^2\right) \geq 1 - \eta$$

uniformly over the mean vector and all variance vectors satisfying the assumption. Replacing each contrast standard error by $\widehat{\sigma}_{\max} \|w_T\|_2$ and using error budget $\alpha - \eta$ therefore gives at least $1 - \alpha$ coverage.

The proof uses Anderson's inequality for a translated Gaussian norm [Anderson, 1955] and a weighted chi-squared lower-tail inequality [Laurent and Massart, 2000]. The method is a sensitivity analysis: its claim is only as credible as the chosen κ. Larger values correctly give wider sets.

6.3 Independent replicate curves

Suppose we observe R independent curves

$$Y_r \sim N(\mu, \Sigma), \qquad r = 1, \ldots, R,$$

where the common covariance Σ may be non-diagonal or singular. For a contrast row w_j, let $Z_{rj} = w_j^\top Y_r$, with sample mean $\bar{Z}_j$ and sample standard deviation s_j across replicates.

Theorem 6.4 (Replicate Student band)**.** *For M fixed contrasts, the intervals*

$$\bar{Z}_j \ \pm \ t_{R-1,1-\alpha/(2M)} \frac{s_j}{\sqrt{R}}, \qquad j = 1, \ldots, M,$$

cover all contrast means simultaneously with probability at least $1 - \alpha$. Their sign inversion therefore satisfies $\mathrm{P}\{\mathcal{I}(\mu) \subseteq \widehat{\mathcal{C}}\} \geq 1 - \alpha$.

Dependence and unequal variance across design locations are allowed inside one replicate. Independence is required across replicates, which is the correct sampling unit in repeated-curve studies [Davidian and Giltinan, 1995].

7 Computation and matched honest baseline

7.1 Matrix-free implementation

Writing all contrast rows as an $M \times n$ matrix is unnecessary. On a uniform design, block averages and chord residuals can be evaluated from prefix sums. The implementation stores only scale metadata and start indices. The default separation family uses ratios $(1, 2, 4)$ across dyadic scales.

Table 1 gives a deterministic scaling benchmark. Times are hardware-specific, while storage and dense counterfactual sizes follow directly from the constructed family. At $n = 10^6$, the implementation evaluates 5,999,750 contrasts in 0.73 seconds and stores 53.4 MiB. A dense double matrix would require about 44,702 GiB.

7.2 Pointwise-band projection

To compare two methods with the same type of guarantee, we construct a generic pointwise confidence box

$$\mathcal{B}(Y) = \prod_{i=1}^{n} [Y_i - z_{1-\alpha/(2n)}\sigma, Y_i + z_{1-\alpha/(2n)}\sigma].$$

Table 1: Matrix-free scaling for the full $(1, 2, 4)$ separation family.

n	Contrasts	Build (s)	Evaluate (s)	Stored (MiB)	Dense (GiB)
1,000	5,872	0.0006	0.0015	0.05	0.04
10,000	59,831	0.0009	0.0039	0.53	4.46
100,000	599,790	0.0042	0.0287	5.34	446.88
1,000,000	5,999,750	0.0346	0.7318	53.40	44,701.62

For each possible cut, a linear program asks whether this box contains a vector that is discretely convex before the cut and concave after it. The outer range of feasible cuts is PBP. If the complete mean vector lies in the box, every true cut is feasible. Bonferroni therefore gives the same finite-sample coverage target as SCI. PBP spends error probability on every coordinate; SCI spends it on the feature-specific contrasts. The comparison measures the gain from targeting the scientific feature directly.

8 Simulation experiments

8.1 Design

We use four benchmark signals from the S-shaped estimation study of Feng et al. [2022]: a cusp, a one-sided onset, a jump, and a smooth logistic curve. The first three stress nonsmooth behavior; the logistic signal is the smooth reference case. Designs are either uniform or sampled from a Beta$(4, 8)$ distribution. Sample sizes are $n = 500$ and $n = 1000$, the Gaussian standard deviation is 0.1, and the nominal confidence level is 95%.

The study was staged. An initial comparison used 200 responses per cell and included the S-shaped least-squares point estimator (`Sshaped` 1.2) and the residual-bootstrap interval from `ShapeChange` 1.5 with 1,000 bootstrap samples. We then froze the SCI configuration and ran a fresh high-precision audit with 5,000 responses in each of 16 cells, or 80,000 responses in total. Finally, a separate 200-response-per-cell experiment compared SCI with the matched honest PBP baseline. Failures and empty sets were never discarded.

8.2 High-precision coverage audit

Table 2 summarizes the 80,000-response audit. Coverage is stable between 0.9770 and 0.9804. The simultaneous contrast band itself covers all contrast means in 0.9524–0.9618 of responses, close to the intended 0.95 familywise level. The larger transition-set coverage is expected because not every missed contrast bound changes the inverted set.

Table 2: High-precision known-scale audit. Each row summarizes four fixed design–sample-size cells with 5,000 responses per cell.

Signal	Coverage range	Median-width range	Largest empty rate
Cusp	0.9772–0.9804	0.0712–0.1737	0.0182
Onset	0.9772–0.9788	0.6703–0.8149	0.0066
Jump	0.9770–0.9798	0.0020–0.0100	0.0228
Logistic	0.9776–0.9798	0.9990–1.0000	0.0006

The widths contain the main scientific message. The jump is localized very sharply, the cusp moderately, and the one-sided onset only partly. The weak logistic curvature does not produce reliable local signs, so the answer is almost the whole design range. This is not a computational failure. It is the honest statement that the chosen data and noise level do not identify the transition precisely.

One preregistered diagnostic required zero empty sets. It failed: empty-set frequencies reached 2.28%. Theorem 5.1 only requires this probability to be at most 5% for a nonempty target, and the Wilson upper bounds remain below 5% in all cells. We nevertheless report the frozen diagnostic as failed rather than redefining it after seeing the results.

8.3 Published comparator and hybrid workflow

Figure 2 shows the initial comparison. SCI coverage was 0.965–0.995. The `ShapeChange` residual-bootstrap interval had zero coverage in 9 of 16 cells, all involving nonsmooth signals outside its documented smooth spline regime. In the four smooth logistic cells its coverage was 0.960–0.990 and its intervals were narrower. The correct use is therefore a hybrid: retain a strong S-shaped estimator as the point summary, and use SCI when the uncertainty claim must remain valid under nonsmoothness.

The hybrid did not improve point-estimation error. Aggregated mean absolute error was 0.06236 for the published S-shaped estimator and 0.06276 after a simple projection. We therefore make no "point-estimator booster" claim. The value of SCI is calibrated uncertainty around the point estimate.

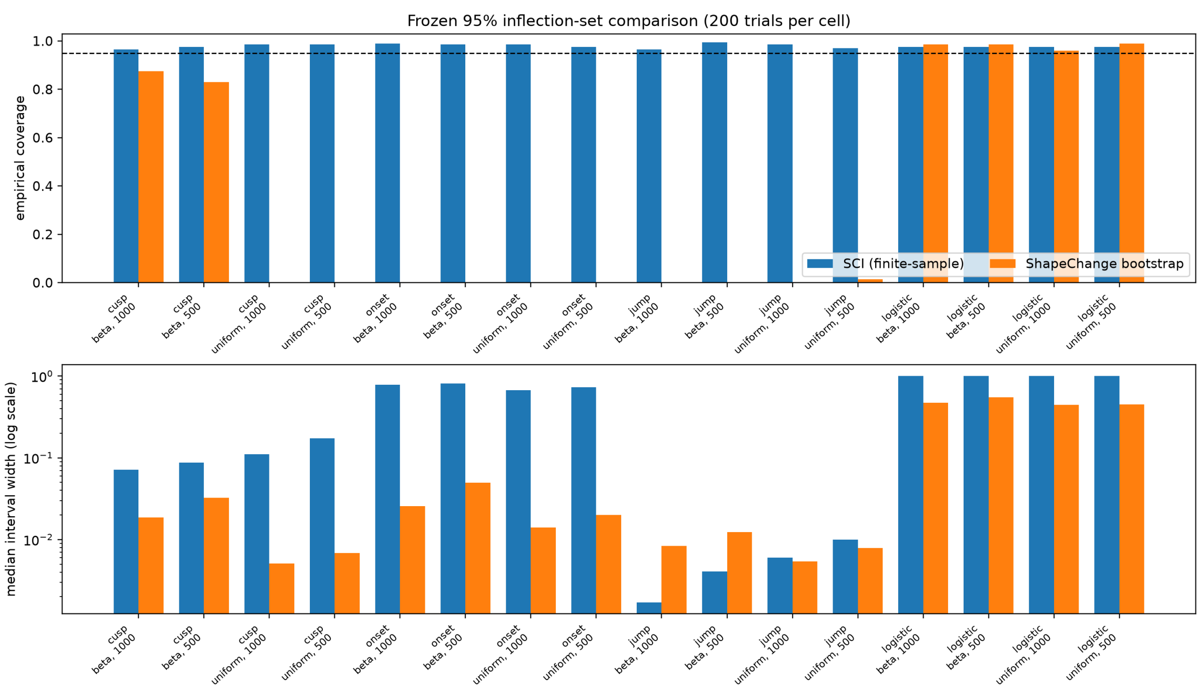

Figure 2: Coverage and width in the initial comparison. The smooth-bootstrap comparator is used inside and outside its smooth regime to expose the practical robustness gap; the comparison does not contradict its published assumptions.

8.4 Matched honest comparison

Table 3 compares SCI and PBP under the same Gaussian model and nominal coverage target. Both methods covered in every cell at the expected or conservative rate. For the cusp, onset, and jump, SCI reduced median width by 19.4%–75.7%. For the weak logistic signal, both methods returned the same nearly full range. This experiment isolates the useful gain: direct feature-specific inversion can be much sharper than projecting a generic confidence box, but it cannot manufacture information when curvature is too weak.

Table 3: Matched honest comparison with 200 responses per cell. "Beta" is the Beta$(4, 8)$ design. Reduction is relative median width.

Signal	Design	n	SCI cov.	PBP cov.	SCI width	PBP width	Reduction
Cusp	Uniform	500	.985	1.000	.1737	.4122	57.9%
Cusp	Uniform	1000	.965	1.000	.1109	.3861	71.3%
Cusp	Beta	500	.950	1.000	.0877	.3151	72.2%
Cusp	Beta	1000	.995	1.000	.0712	.2934	75.7%
Onset	Uniform	500	.975	.995	.7325	.9960	26.5%
Onset	Uniform	1000	.980	1.000	.6703	.9980	32.8%
Onset	Beta	500	.980	1.000	.5460	.6776	19.4%
Onset	Beta	1000	.985	1.000	.5170	.7116	27.3%
Jump	Uniform	500	.965	1.000	.0100	.0180	44.4%
Jump	Uniform	1000	.980	.995	.0050	.0090	44.4%
Jump	Beta	500	.975	.995	.0041	.0068	40.0%
Jump	Beta	1000	.985	1.000	.0017	.0034	50.0%
Logistic	Uniform	500	.970	1.000	.9960	.9960	0.0%
Logistic	Uniform	1000	.965	1.000	.9980	.9980	0.0%
Logistic	Beta	500	.975	1.000	.6776	.6776	0.0%
Logistic	Beta	1000	.975	1.000	.7116	.7116	0.0%

For this matched experiment, both procedures were restricted to the observed design range $[x_1, x_n]$. This matters for the random Beta design: its observed range is shorter than $[0, 1]$. The high-precision audit in Table 2 instead used the full scientific domain $[0, 1]$.

8.5 Unknown-scale and heteroskedastic checks

The unknown-homoskedastic experiment contained 28 frozen cells. The minimum SCI coverage was 0.990, the minimum coverage of the upper scale bound was 0.975, and the mean width inflation among informative cells was 1.117 relative to known scale. All frozen gates passed.

The bounded-heteroskedastic experiment contained 48 cells spanning constant, linear, and plume-like variance profiles. The minimum transition coverage and the minimum scale-envelope coverage were both 0.995. These results test the implementation of Theorems 6.1 and 6.3; the guarantees themselves come from the theorems, not from Monte Carlo performance.

9 Real-data illustrations

9.1 Atmospheric LIDAR

The LIDAR example studies a response curve over altitude. A published S-shaped point fit places the transition at 588 m. The `ShapeChange` fit gives 606 m with a residual-bootstrap interval $[592.6, 612.8]$ m. Visible variance changes make that iid-bootstrap interval descriptive here.

Figure 3 reports the SCI sensitivity analysis. With $\kappa = 1$, the set is $[510, 664]$ m. It grows to $[486, 720]$ m for $\kappa = 1.5$, $[438, 720]$ m for $\kappa = 2$, and $[390, 720]$ m for $\kappa \geq 3$ in the displayed range.

The conclusion is conditional: if the maximum variance is at most κ times the mean variance, the corresponding set has the finite-sample guarantee. The widening shows the cost of making a more cautious variance claim.

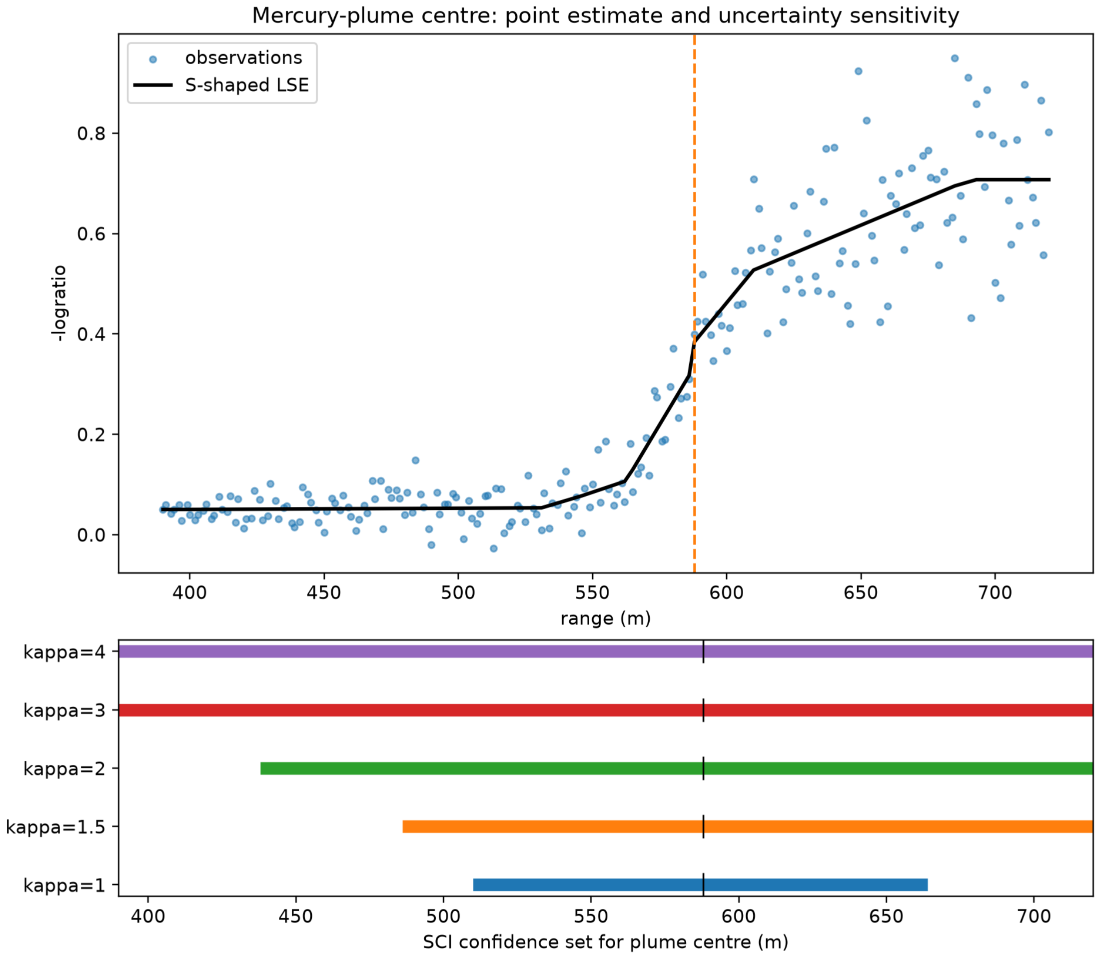

Figure 3: LIDAR transition sets under the declared variance-ratio bound κ. Larger heterogeneity allowances produce wider honest sets.

9.2 Replicated DNase assay

The DNase data distributed with R contain 11 assay runs and eight concentrations [R Core Team, 2025]. We use all 176 raw optical-density measurements. Technical duplicates are averaged within each run, and the run is the independent unit. This leaves 11 replicate curves on the same eight-point design.

The replicate Student construction gives the 95% transition set $[0.78125, 12.5]$ in concentration units. A four-parameter logistic fit gives the descriptive midpoint 4.1412. The point lies inside the confidence set, but the set reaches the largest observed concentration. The data support a lower limit, while they do not supply a reliable upper limit inside the observed range.

The exact statement assumes independent Gaussian replicate curves with a common mean and covariance. With only 11 runs, the interval is necessarily conservative. Treating all 176 readings as independent would give a narrower but invalid answer because technical duplicates and concentrations from the same run are correlated.

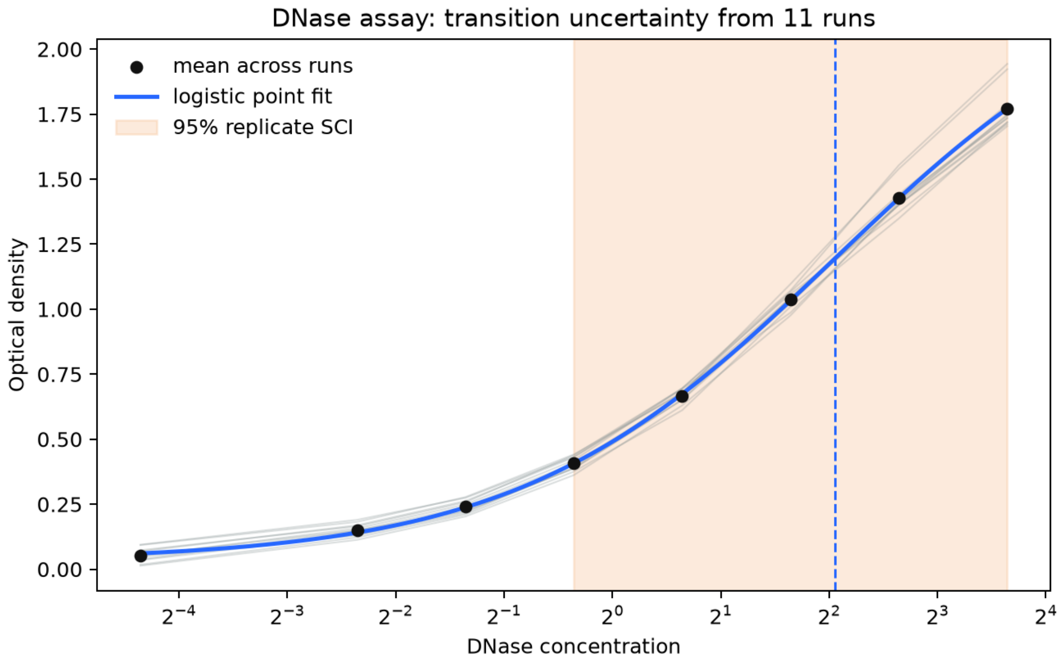

Figure 4: DNase run means, replicate mean curve, descriptive logistic midpoint, and the 95% replicate-curve SCI set. Dependence across concentrations inside a run is allowed.

10 Discussion

10.1 What practical problem is solved?

The method solves a specific uncertainty problem: it gives a finite-sample outer confidence set for every convex-to-concave transition that is compatible with a sampled mean curve, without assuming a smooth or unique inflection. A researcher can attach this set to an S-shaped point estimate and distinguish three outcomes:

1. a narrow set, meaning that the data certify local shape signs close to the fitted transition;
2. a one-sided or wide set, meaning that only part of the location is identified; and
3. an empty set, which is a controlled rare event under the model but can also flag model tension in practice.

This addresses the gap exposed by the experiments. A smooth bootstrap can be sharp when its assumptions hold but fail badly on nonsmooth signals. A generic honest confidence region remains valid but can be unnecessarily wide. SCI keeps the broad nonsmooth guarantee while using feature-specific evidence to improve width in informative cases.

10.2 What is not claimed?

First, SCI is not a new point estimator and did not improve point-estimation error in our hybrid experiment. Second, it is not the first inference method for inflection points. Third, the $(\log n/n)^{1/7}$

cubic rate is not new. Fourth, the output is an outer set, not a proof that every included point is a true physical threshold. Finally, the real-data conclusions are conditional on their sampling models.

10.3 Limitations and open work

The exact single-curve results are Gaussian. A self-normalized or distribution-robust band would broaden the method. The heteroskedastic extension needs a user-declared κ; it cannot learn an unrestricted variance vector from one observation per design point. The replicate result needs independent runs with a common covariance. Bonferroni calibration is simple and transparent but can be conservative when many contrasts overlap.

The main theoretical gap is also clear. Theorem 5.4 gives a rate under explicit contrast-margin and availability conditions. A broad function-class lemma that derives these conditions automatically, together with a matching lower bound for the set-valued target, remains open. Such a result could turn the current conditional rate into a minimax characterization.

11 Conclusion

Shape-contrast inversion turns local geometric evidence into an honest uncertainty statement for a convex-to-concave transition. Its main strength is not a universally narrow interval. It is the combination of finite-sample coverage, tolerance of nonsmooth and nonunique transitions, and a direct matrix-free implementation. The matched comparison shows that targeting the feature can be much sharper than projecting a generic confidence box. The logistic and DNase examples show the equally important other side: when the data do not support precise localization, the method says so.

Reproducibility statement

The public repository contains the matrix-free implementation, frozen experiment configurations, raw trial-level outputs, summaries, figures, and verification manifests. All reported numerical claims are generated from these artifacts. The release test suite contains 228 tests, including 48 specific to SCI. The release manifest verifies 1,565 files, and the SCI artifact verifier checks 60 frozen SCI files. Commands for rebuilding this manuscript are given beside the source.

CRediT authorship contribution statement

Saveliy Baturin: Conceptualization; Methodology; Software; Validation; Formal analysis; Investigation; Data curation; Visualization; Writing – original draft; Writing – review and editing.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported here.

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The author used OpenAI Codex to assist with literature organization, software engineering, verification, and language editing. The mathematical claims, experimental design, reported results, and final interpretations were checked and approved by the author, who assumes full responsibility for the manuscript.

A Contrast construction

The implementation uses a fixed dyadic family. For a scale q, a start index s, and separation multiplier $g \in \{1, 2, 4\}$, it selects left, middle, and right blocks of comparable size with separation gq. Each block chord residual is a nonnegative average of elementary terms of the form (1). The exact weights account for irregular design spacing. Their sum is zero, so affine functions have zero contrast. Convex functions give a nonnegative mean and concave functions a nonpositive mean.

The family includes all admissible starts at each dyadic scale. It is fixed from the design before the responses are read. On a uniform design, prefix sums of Y_i and iY_i evaluate each block average in constant time. The known-scale standard error is computed from the closed-form squared norm of the stored row. Irregular designs use sparse index and weight arrays rather than a dense $M \times n$ matrix.

B Proofs

Proof of Lemma 3.2. Because $m_2 \in \mathcal{I}(f)$, the function is convex on $[A, m_2]$, hence on $[m_1, m_2]$. Because $m_1 \in \mathcal{I}(f)$, it is concave on $[m_1, B]$, hence on the same interval. A real-valued function that is both convex and concave on an interval is affine there. For any $m \in [m_1, m_2]$, membership of m_2 gives convexity on $[A, m]$, while membership of m_1 gives concavity on $[m, B]$. Thus $m \in \mathcal{I}(f)$. □

Proof of Theorem 5.1. For each T, $Z_T - \mu_T$ is centered normal with standard deviation $\sigma \|w_T\|_2$. By (2) and a union bound,

$$\mathrm{P}\{\ell_T \le \mu_T \le u_T \text{ for all } T \in \mathcal{T}\} \ge 1 - \alpha.$$

Call this event E and fix $m \in \mathcal{I}(f)$. If $\ell_T > 0$ and $a_T \ge m$, the support lies in the concave part of f, so $\mu_T \le 0$. This contradicts $\ell_T \le \mu_T$ on E. Therefore every certified positive contrast has $a_T < m$, and $\widehat{L} \le m$. Likewise, if $u_T < 0$ and $b_T \le m$, the support lies in the convex part, so $\mu_T \ge 0$, contradicting $\mu_T \le u_T$. Hence $m \le \widehat{U}$. The argument holds for every $m \in \mathcal{I}(f)$ on the same event E, so $\mathcal{I}(f) \subseteq [\widehat{L}, \widehat{U}] = \widehat{\mathcal{C}}(Y)$. The uniform bound follows because the probability of E does not depend on f. □

Proof of Proposition 5.2. A certified positive contrast rules out only points at or below its left endpoint by the stated sign logic. Combining all such statements gives the largest lower limit $\widehat{L}$. A certified negative contrast rules out only points at or above its right endpoint, giving the smallest upper limit $\widehat{U}$. Thus every point in $[\widehat{L}, \widehat{U}]$ survives all certified implications, while every point outside violates at least one of them. Any closed interval based only on these implications and containing all surviving points must contain $[\widehat{L}, \widehat{U}]$. □

Proof of Theorem 5.4. The simultaneous band event from Theorem 5.1 has probability at least $1 - \alpha$. Separately,

$$\mathrm{P}\left\{Z_{T_-} > \mu_{T_-} - z_{1-\eta/2}\sigma \left\|w_{T_-}\right\|_2\right\} \ge 1 - \eta/2,$$

and the analogous upper-tail statement holds for T_+. A union bound gives all three events with probability at least $1 - \alpha - \eta$. Condition (7) then implies $\ell_{T_-} > 0$ and $u_{T_+} < 0$. Therefore $\widehat{L} \ge a_{T_-} \ge m_- - Kd$ and $\widehat{U} \le b_{T_+} \le m_+ + Kd$. The simultaneous band event also gives $\mathcal{I}(f) \subseteq \widehat{\mathcal{C}}(Y)$ by Theorem 5.1. The length bound follows. □

Proof of Corollary 5.5. For a symmetric equally spaced triple with middle location u and physical separation d, direct expansion of the cubic gives

$$Q(f) = 3B(m_0 - u)d^2. \tag{13}$$

Take an implemented multiplier-one row of block size q. Its q triples have physical separation $d = q/(n+1)$ and use disjoint left, middle, and right blocks. Its coefficient norm is therefore exactly

$$\|w_T\|_2 = \sqrt{\frac{3}{2q}}. \tag{14}$$

Choose the rightmost allowed start, on the q-spaced grid of starts, whose whole support ends at or before m_0. The interior condition ensures that it exists. Every triple has its right point at or before m_0, so its middle point is at least d to the left of m_0. Equation (13) gives $\mu_T \geq 3Bd^3$. Maximality of the start also puts its support beginning less than $4d$ to the left of m_0. With (14),

$$\frac{\mu_T}{\sigma \|w_T\|_2} \geq \frac{\sqrt{6}B\sqrt{n+1}\, d^{7/2}}{\sigma} \geq c_{\alpha,M_n} + z_{1-\eta/2},$$

where the last step uses $q \geq q_*$. This row fails to be certified positive with probability at most $\eta/2$.

The symmetric construction uses the leftmost q-grid start whose support begins at or after m_0. It gives a negative row with failure probability at most $\eta/2$ and support ending less than $4d$ to the right of m_0. When both rows are certified, $\widehat{L} > m_0 - 4d$ and $\widehat{U} < m_0 + 4d$. The smallest-dyadic choice gives $q < 2q_*$, so $d < 2h_*$ and $\widehat{U} - \widehat{L} < 16h_*$. Intersecting with the simultaneous-band event of Theorem 5.1 retains m_0 and gives total probability at least $1 - \alpha - \eta$.

At each dyadic q and separation multiplier, there are at most $n/q + 2$ starts. Summing n/q over dyadic sizes gives less than $2n$. With three multipliers and at most $1 + \log_2 n$ sizes, $M_n \leq 6n + 6(1 + \log_2 n)$. Thus $c_{\alpha,M_n} = O(\sqrt{\log n})$, which gives the stated order. □

Proof of Theorem 6.1. Write $R_X Y = R_X\mu + \sigma R_X G$ with $G \sim N(0, I)$. The set $\{v : \|v\|_2^2 \leq q\}$ is convex and centrally symmetric in the residual subspace. Anderson's inequality implies that translating a centered Gaussian cannot increase its probability of lying in this set. Hence

$$\mathop{\mathrm{P}}_{\mu,\sigma}\left\{\frac{\|R_X Y\|_2^2}{\sigma^2} \leq q_{n-r,\eta}\right\} \leq \mathrm{P}\{\chi^2_{n-r} \leq q_{n-r,\eta}\} = \eta.$$

Thus $\overline{\sigma} \geq \sigma$ with probability at least $1 - \eta$. On this event, bands using $\overline{\sigma}$ are no narrower than valid bands using σ. Combining this event with a contrast-band error budget $\alpha - \eta$ proves the result by a union bound. □

Proof sketch of Theorem 6.3. Let $D = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2)$. Anderson's inequality first removes the unknown translated mean from the lower tail of $Q = \|RY\|_2^2$. Diagonalizing $D^{1/2}RD^{1/2}$ writes the centered quadratic form as $\sum_j \lambda_j G_j^2$. The block construction gives the needed lower bound on $\sum_j \lambda_j$; Assumption 6.2 bounds $\max_j \lambda_j$ and $\sum_j \lambda_j^2$. The weighted chi-squared lower-tail inequality of Laurent and Massart [2000] then gives

$$Q \geq \frac{n\bar{\sigma}^2}{2} d_{n,\kappa,\eta}$$

with probability at least $1 - \eta$. Multiplication by $2\kappa/(nd_{n,\kappa,\eta})$ and $\max_i \sigma_i^2 \leq \kappa\bar{\sigma}^2$ proves the scale envelope. The SCI coverage statement follows by the same error-budget union bound as in Theorem 6.1. □

Proof of Theorem 6.4. For each fixed j, the replicate contrasts $Z_{1j}, \dots, Z_{Rj}$ are iid univariate Gaussian. The usual Student statistic therefore has a t_{R-1} distribution whenever its variance is positive. In a zero-variance direction the sample mean equals the contrast mean almost surely, so the degenerate interval is exact. A Bonferroni union bound over the M two-sided intervals gives simultaneous coverage at least $1 - \alpha$. The deterministic sign-inversion argument from Theorem 5.1 completes the proof. $\square$

C Additional experimental detail

All stochastic experiments use recorded seeds and frozen JSON configurations. The high-precision audit was run independently of the development simulations. For each cell, coverage is the fraction of responses whose SCI set contains the known transition target. Empty sets count as noncoverage. Width is zero for an empty set and otherwise is the interval length; both the empty frequency and median width are therefore reported.

The PBP feasibility problem uses the same observed design and shape definition as SCI. For a candidate cut k, the decision variables are a mean vector $\theta \in \mathbb{R}^n$ constrained to the pointwise confidence box. Divided slopes of θ are required to be nondecreasing before k and nonincreasing after k. Feasibility is monotone on each side of the outer range, which permits a binary search over candidate cuts. This baseline is exact for the sampled shape constraints and intentionally makes no smoothing assumption.

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